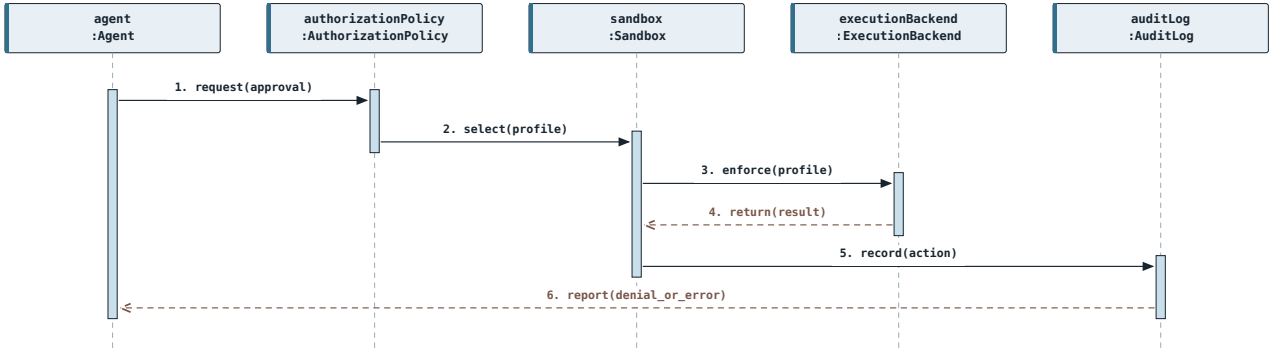
P23 Layered permission and sandbox

Primitive / OBJECT INTERACTION / ANALYTICAL COLLABORATION

Combine policy decisions with execution isolation.

OBJECT INTERACTION

solid arrow: call/message · dashed open arrow: return/failure · analytical order, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	A tool action is evaluated by policy and executed in a controlled environment.
TRIGGER	A privileged operation or command is dispatched.
STRUCTURE	Authorization decision, sandbox profile, environment adapter, and enforcement point.
EFFECT	The effective capability equals the authorized capability, including filesystem and network.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	Requested operation and active environment policy are both observable.
ENSURES	Any difference between approval and enforcement fails closed or asks for escalation.
INVARIANT	A policy decision cannot be bypassed by switching sandbox or transport backend.
VIOLATION	Enforcement errors deny execution and preserve actionable diagnostics.

Sources: protocol/pattern_diagram_models.yaml (participants + messages); protocol/harness_pattern_specification.yaml (pointcut + contract).

Analytical collaboration only; no repository execution or message order is asserted.